



Chewbacca tribute in space

NASA astronaut paid a tribute to the late Chewbacca actor, Peter Mayhew, while in space. <https://dgit.in/june19-24>



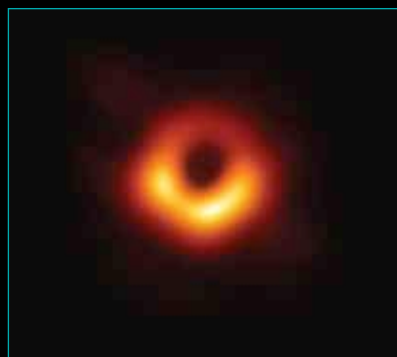
Astroid destroys NYC...

In a simulation! NASA destroyed New York City in an asteroid strike simulation. <https://dgit.in/june19-24>

Way, and has the designation of Sagittarius A* (don't go looking for a footnote, the name is read as Sagittarius A-star). Sagittarius A* is at a distance of 26,000 light years. However, this target is difficult to observe because of the light pollution from all the stars between the Earth and Sagittarius A*. One of the most massive black holes known, with a mass of 6.5 billion Suns is the black hole at the centre of the M87 galaxy, 53 million light years away. The object had been imaged in the infrared and near infrared spectrum by the Hubble Space telescope in 1998, clearly capturing the astrophysical jet emanating from M87*. This is a beam of ionised matter that is perpendicular to the plane of the accretion disk, or the matter falling inwards into the black hole. That jet, which can be seen in the cover image of this article, stretches for a staggering 5000 light years. The particles in the jet have high enough energy to travel at close to the speed of light. Sagittarius A* and M87* were the twin targets of an ambitious project that was launched in 2009, to capture a direct image of the event horizon of a black hole.

The Event Horizon telescope is not a single astronomical instrument, but a virtual telescope that combines the data of multiple telescopes around the world. Every telescope has an aperture, or a distance over which it captures data. The wider the aperture, the more information is captured, and the higher resolution photos you get. The Event Horizon Telescope was formed using the Atacama Large Millimeter Array (ALMA) in Chile and Atacama Pathfinder Experiment in Chile, the IRAM 30-metre telescope in Spain, the James Clerk Maxwell Telescope in Hawaii, the Large Millimeter Telescope Alfonso Serrano in Mexico, the Submillimeter Array as well as the Submillimeter Telescope in the USA, and the South Pole Telescope. Out of these two, the ALMA and the Submillimeter Array are themselves virtual telescopes. With the data of all these telescopes

combined together, the resulting aperture was roughly the same as the diameter of the Earth. In April 2017, all the eight telescopes started observing M87* at the same time, over a period of ten days. The technique used for capturing the data is known as very-long-baseline interferometry or VLBI. The data from each telescope was recorded, time stamped and synchronised using atomic clocks. Each of the eight telescopes generated around 350 terabytes every day. A total of 5 petabytes of data was collected, and as the internet cannot yet handle such loads, the data was stored on helium filled hard



The first ever direct visual evidence of a supermassive black hole and its shadow

disks, and flown to supercomputers called "corellators" at the Max Planck Institute for Radio Astronomy in Germany and MIT Haystack Observatory in the US.

THE SHADOW OF A BLACK HOLE

After a decade of coordination, a team of over 200 scientists around the world managed to capture the first ever image of a black hole and its shadow. The more massive a black hole, the larger its shadow is. For this reason, M87* actually made a great target for scientists who wanted to achieve a very specific breakthrough in astrophysics - capturing direct evidence of the shadow of a black hole. Calling this achievement the "first photo of a black hole" is a simplistic and not entirely accurate descrip-

tion. Astrophysical jets, gravitational waves, and even the flares associated with black holes capturing stars had been observed before, but not the boundary beyond the accumulation of superhot gases around a black hole. Einstein's theory of general relativity predicts that black holes will have shadows. These are dark regions of space caused by the light falling into the event horizon. The event horizon itself is 2.5 times smaller than the shadow. The image was compared to computer simulations of how the object would appear, based on contemporary understandings of how the universe works. These simulations were created using 140 Nvidia GPUs, and matched closely with the observations. Paul T.P. Ho, EHT board member said, "Once we were sure we had imaged the shadow, we could compare our observations to extensive computer models that include the physics of warped space, superheated matter and strong magnetic fields. Many of the features of the observed image match our theoretical understanding surprisingly well. This makes us confident about the interpretation of our observations, including our estimation of the black hole's mass."

Not too many details are visible in the image, but it helps us understand the incredible phenomena that takes place in and around a black hole. The darker regions represent matter moving away from the Earth, while the lighter regions represent matter moving towards the Earth. Six scientific papers were published to mark the breakthrough, and appeared in a special issue of The Astrophysical Journal Letters.

EHT project director Sheperd S. Doeleman said, "We have achieved something presumed to be impossible just a generation ago. Breakthroughs in technology, connections between the world's best radio observatories, and innovative algorithms all came together to open an entirely new window on black holes and the event horizon." 